signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

*signal.sci

```
// Time vector
t = -1:0.001:1; // fs = 1000 Hz
x = exp(-t.^2); // Gaussian pulse
// FFT length (must be >= length(x))
N = 2048; // choose bigger than length(x)
Xf = fft([x zeros(1, N - length(x))]); // zero-padding to N
// Frequency vector
fs = 1000; // Hz
f = (-N/2:N/2-1) * (fs/N);
// Magnitude spectrum
magX = abs(fftshift(Xf));
// Phase spectrum
phaseX = atan(imag(fftshift(Xf)), real(fftshift(Xf)));
// Plot

clf;
subplot(2,1,1);
plot(f, magX);
xtitle("Magnitude Spectrum", "Frequency (Hz)", "Magnitude");
subplot(2,1,2);
plot(f, phaseX);
xtitle("Phase Spectrum", "Frequency (Hz)", "Phase (rad)");

```

Line 24, Column 0.

Graphic window number 0

File Tools Edit ?

Graphic window number 0 ?

